SIMATSENGINEERING(SAVEETHASCHOOLOFENGINEERING)

DepartmentofArtificialIntelligenceandDataScience ASSESSMENTTOOL2 – CaseStudy

CourseCode:	DSA03	CourseTitle:	NaturalLanguageProcessing
COAssessed:	CO3 – Precision	Assessment:	AssessmentTool2– Cas
Weightage:	40% of		
CO5Statement:	ApplyN-grammodels,smoothingtechniques,part-of-speechtaggingsystemsandt oevaluate effective syntactic analysis. (BL-3)		
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Section/Batch:	B.techAI-ML	Date:	08-08-2026

CodeofConduct

I Y. Harsha Vardhan Reddy (Name /Reg No) certify that thissubmission is myoriginalworkand thatIhaveadheredtotheguidelinesspecifiedforthisassessment.Iunderstandthatanyviolationofacademicintegrity rules will result in disciplinary action.

Signature of theStudent:

Instructions

- Readallthreecasestudiescarefullybeforeattemptingthequestions.
- Provideappropriatejustificationsandinterpretationsforeverycomputedresult.
- UseNLPconcepts,algorithms,andterminologywhereverapplicable.
- Diagrams,flowcharts,tables,ormathematicalformulationsmaybeusedtosupportyouranswers. • Duration:30minutes.

N-Gram Language ModelsandSmoothingTechn	Bigram and Trigram probability computation,Maximu Likelihood Estimation (MLE), Backoff models, Deleted Interpolation, Entropy calculation, uncertainty analysis, real-time next-word	CaseStudy1–Smart Mob Keyboard PredictionSy prediction
Part-of-Speech (POS)TaggingandWordClas	Penn Treebank tagset, contextual ambiguityresolution, rule-based tagging, stochastic (HMM), emission and transition probabilities, wordclassidentification,chatbotlanguageunderstandi g	CaseStudy2–AI-PoweredCustomer tagging Support Chatbot
Transformation-Based Tagging (Brill Tagg	POStagcorrection,transformationrules,contextualtag linguistic validation, tag sequence refinement, ambiguity handling	Case Study 3–AnalyticsandPOSTag Correction System
CorpusStatisticsand ProbabilisticAnalysis	Word frequency counting, corpus-based estimation, entropy-based uncerta measurement, tagging confidence evaluat	Case S probability AnalyticsandPOSTag Correction Syst

IntegratedSyntactic Analysis and NLPApplicatio	Application of language models, POSprobabilistic methods, corpus analysis, ambiguity resolution, and decision-making in real-\ systems	CaseStudies1,2,and3 tagging, NLP

News

News

CASESTUDY1:SMARTMOBILEKEYBOARDPREDICTIONSYSTEM

Introduction

An AI-powered mobile keyboard predicts the next word based on previously typedwords. N-gram language models such as Bigram and Trigram models estimate the probabilityofupcoming words using word sequences observed in a training corpus. Backoff and interpolation techniques help the system handle unseen or rare sequences, whileentropymeasures the uncertainty associated with predictions.

GivenCorpus:

“Datascienceispowerful,datasciencedrivesinnovation,datascienceisevolving.” Given Statistics:

C(data)=3

C(data science) =

3 C(science is) = 2

C(sciencedrives)=1

C(sciencedrivesinnovation)=1

InterpolationWeights:

λ_1 = 0.5 –

Trigram λ_2 = 0.3

– Bigram λ_3 =0.2

– Unigram

PredictionProbabilities:

P(is) =0.66

P(drives)=0.33

1. MaximumLikelihoodEstimateofBigramProbability

TheMaximumLikelihoodEstimateof abigramiscalculated using: $P(w_2 | w_1) = C(w_1, w_2) / C(w_1)$

Forthebigram“datascience”:

P(science|data)=C(datascience)/C(data)

Given:

C(datascience)=3

C(data) = 3

Therefore,

P(science|data)=3/3

P(science | data) = 1.0

Interpretation

Theprobabilityof1.0meansthatinthegiventrainingcorpus,everyoccurrenceoftheword“data”isf
ollowed by the word “science”.
Therefore,whenausertypes:
data →?
themobilekeyboardwillstrongly predict:
data →science
This demonstrates how bigram probabilities allow predictive keyboards to suggest themostlikely next word based on the immediately preceding word. However, the probabilityisbasedonly on the available training corpus, so a larger and more diverse corpus is requiredforrealistic applications.
2. BackoffModelforanUnseenSequence

Considertheunseensequence:
“data science improves”
Thetrigramprobability is:
P(improves|data science)
Thetrigram “datascienceimproves”doesnot occur inthe given corpus. Therefore,
C(datascienceimproves)=0
anddirecttrigram estimationcannotprovideauseful probability.

ApplyingBackoff
Themodelfirstattemptsthetrigram:
P(improves|data science)
Sinceitisunseen, themodelbacksofftothebigram:
P(improves|science)
Thebigram“scienceimproves”isalsonotpresent inthe corpus.
Therefore,themodelbacksoffagaintothe unigram:
P(improves)
If“improves”isalsoabsentfromthetrainingvocabulary,itsunsmoothedunigramprobabilityis:
P(improves) = 0
Hence,usingthesuppliedcorpusalone:
P(datascienceimproves)=0

In a practical language model, smoothing or an <UNK> unknown-word probability wouldnormally be used so that the sequence receives a small non-zero probability.

Importanceof Backoff
Backoff is essential because users frequently type word combinations that are not presentinthe training dataset. It allows the system to move from a specific higher-order model toamoregeneral lower-order model.

Forexample:
Trigram→Bigram→Unigram
Thisimprovescoverage,preventsthepredictionmechanismfromfailingonunseensequences,and supports real-time prediction even when the available language data is sparse.

3. DeletedInterpolationfor“datascienceis”
DeletedinterpolationcombinesTrigram,Bigram,andUnigramprobabilities. The general formula is:
$P(w_3|w_1,w_2)=\lambda_1 P(w_3|w_1,w_2)+\lambda_2 P(w_3 \mid w_2)+ \lambda_3 P(w_3)$
Given:
$\lambda_1 = 0.5$
$\lambda_2 = 0.3$
$\lambda_3 = 0.2$
Forthesequence:

datascienceis

Step1:TrigramProbability
The corpus
contains:“datascienceis”
=2times and:
“datascience”=3 times
Therefore,
P(is|data science)=2/3
P(is|datascience)=0.667

Step2:Bigram Probability
C(scienceis)=2
Theword “science”occurs3 times.
P(is|science)=2/ 3
P(is|science)=0.667

Step3:UnigramProbability
Ignoringpunctuation,thecorpuscontains12word tokens:
Datascienceispowerful
datasciencedrivesinnovatio
n data science is evolving
Theword“is” occurs2 times.
P(is) =2 / 12
P(is)= 0.167

Step4:ApplyInterpolation
P(is|datascience)
=(0.5×0.667)+(0.3×0.667)+(0.2×0.167)
=0.3335 +0.2001 +0.0334
P(is|datascience)≈0.567

Interpretation
Theinterpolatedprobabilityisapproximately:
0.567 or56.7%
InterpolationisusefulbecauseitdoesnotdependcompletelyonasingleN-gramlevel.Itcombines information from multiple models.

The Trigram model provides more contextual information, the Bigrammodel providesbettercoverage whentrigramevidenceis limited,andtheUnigrammodelprovidesgeneralword frequencyinformation. Therefore, interpolation achieves a balance between reliabilityandcoverage, especially when the training dataset is sparse.

4.EntropyAnalysis
Given:
P(is) =0.66
P(drives)=0.33
Entropyiscalculatedusing:
$H(X)=-\Sigma P(x)\log_2 P(x)$
Therefore,
$H=-[0.66 \log_2(0.66) +0.33\log_2(0.33)]$
Approximately,
$\log_2(0.66)\approx -0.599$
$\log_2(0.33)\approx -1.599$

Therefore,
H≈ -[(0.66×-0.599)+(0.33× -1.599)]
H≈ 0.395 +0.528
H≈ 0.923 bits
Thesuppliedprobabilitiestotal0.99becausethey areroundedvalues,sothisisanapproximateentropy.
Interpretation
Anentropyofapproximately0.92bitsindicatesmoderateuncertaintyinthenext-wordprediction. The model prefers:
science→is (0.66)
over:
science→drives(0.33)
However, the probability of “drives” is still significant. Therefore, the model is not completelycertainabout its prediction.

Lowerentropyindicateshigherpredictionconfidence,whereashigherentropyindicatesgreateru ncertainty.
In intelligent keyboards, entropy can help determine whether the systemshould confidentlydisplayone prediction or provide multiple possible word suggestions.

Conclusion
The mobile keyboard prediction system demonstrates how Bigram and Trigrammodelspredictwords from context. MLE calculates probabilities from observed frequencies, Backoff handlesunseen sequences, Deleted Interpolation combines multiple N-gramlevels, andEntropymeasures prediction uncertainty. Together, these techniques improvethe reliabilityandcoverage of intelligent predictive text systems.

CASESTUDY2:AI-POWEREDCUSTOMERSUPPORTCHATBOT

Introduction
Part-of-Speech tagging assigns grammatical categories such as nouns, verbs, adjectives, andadverbs to words in a sentence. In a customer support chatbot, POS tagging helps thesystemunderstand sentencestructureandresolveambiguouswords.HiddenMarkovModelsusetransitionandemissionprobabilities to select the most likely POS tag according to context.

1. PennTreebankPOSTagging
Sentence1
“Bookaflightticketnow.”
POS Tags:
Book/VB
a/DT
flight/NN
ticket/NN
now/RB
TaggedSentence:
Book/VBa/DTflight/NNticket/NNnow/RB

Explanation
“Book”istaggedasVB(Verb)becauseitappearsatthebeginningofanimperativesentenceand represents the action of making a reservation.
Othertagsare:
DT–Determiner
NN – Noun
RB– Adverb

Sentence2
“Thisbookisinteresting.”
POS Tags:
This/DT
book/NN
is/VBZ
interesting/J
J
TaggedSentence:
This/DTbook/NNis/VBZinteresting/JJ

Explanation
Inthesecondsentence,“book”representsanobjectorthing.Therefore,itfunctionsasa Noun(NN). Thus, the same word has different grammatical functions depending on its context: Bookaflight→Book=Verb(VB)
This
book → book = Noun (NN)
ThisdemonstrateslexicalambiguityinNaturalLanguageProcessing.

2. HMMProbabilityfor“book”asaVerb

Given:
P(book | VB) = 0.6
P(book | NN) = 0.4
P(Start→VB)=0.5
P(Start→NN)=0.5
For Verb:
P(VB,book)=P(Start →VB) ×P(book|VB)
=0.5 ×0.6
= 0.30
Forcomparison,asaNoun:
P(NN,book)=P(Start→NN)×P(book |NN)
=0.5 ×0.4
= 0.20
Therefore:
Verb likelihood =
0.30
Nounlikelihood=0.20
Since:
0.30 > 0.20
theHMMfavorstheVB(Verb)tag.

Interpretation
Theword“book”hasagreateremissionprobabilityundertheVBstatethantheNNstate.Sincebothsta
rting transition probabilities are equal, the VB path receives the higher likelihood.
Thegrammaticalcontextalsosupportstheresultbecause“Bookaflightticketnow”isanimperativeco
mmand where “book” expresses an action.

3. Rule-BasedTaggingvsStochasticHMMTagging
Rule-Based Tagging
Rule-basedtaggingusesmanuallydefinedlinguisticrules.
Example:
“If‘book’occursafteradeterminer,tagitasNN.”
Advantages:

•Easytounderstandandinterpret.
•Linguisticknowledgecan bedirectly incorporated.
•Effectiveforwell-definedgrammaticalpatterns.
•Doesnotnecessarilyrequirealargeannotatedcorpus.
Limitations:
•Creatingandmaintainingmanyrulesisdifficult.
•Rulesmayfailforpreviouslyunseensentencestructures.
•Large-scalesystemsrequireaverylargenumber ofrules.

StochasticHMMTagging
HMMtaggingusesprobabilitieslearnedfromtrainingdata.
It considers:

TransitionProbability:ProbabilityofonePOStagfollowinganother.
EmissionProbability:ProbabilityofawordoccurringwithaparticularPOStag.
Advantages:
•Learnspatternsfromlargedatasets.
•Handlesambiguouswordsprobabilistically.
•Canprocesslargeamountsoftext efficiently.
•Providessystematicdecisionsforunseencontexts.
Limitations:
•Requiresannotatedtrainingdata.
•Performancedependsonthequalityandquantityoftrainingdata.
•BasicHMMassumptionsmaynotcapturealllong-rangelinguisticdependencies.

RecommendedApproach
For a large-scale customer support chatbot, stochastic tagging is generally more suitablethana purely rule-basedsystembecauseitcanlearnfromlargedatasetsandhandlelexicalambiguityprobabilistically. Ahybridsystemcanbeevenmoreeffective,wherestatisticalmodelsmaketheprimarypredictionswhi le carefully designed linguistic rules handle specific exceptional cases.

4.RoleofStandardizedPOSTagsetsandWordClasses
StandardizedPOStagsetssuchasthePennTreebanktagsetprovideconsistentgrammaticallabelsf or words.
Examplesinclude:
NN–Noun
VB – Verb
JJ – Adjective
RB –
AdverbDT–Det
erminer
VBZ– Verb,third-person singularpresent

Importancein IntentDetection
POStagshelpidentifyimportantactionsandentities.
For example:
“Bookaflight.”
TheVBtagidentifiesbookasanaction,whileNNtagscanhelpidentifyentitiessuchas flightandticket.
Thishelpsthechatbotidentifythat theuserintendstomake abooking.

ImportanceinResponseGeneration
Grammaticalinformationhelpsthechatbotgenerateresponseswithappropriatesentencestructure and word forms.
HandlingAmbiguity

Wordssuchas“book”canfunctionasmultiplewordclasses.POStagginghelpsthechatbotdistinguish between:
book/NN→anobject
and
book/VB→anaction

OverallPerformance
Standardized tagsets make linguistic analysis consistent across large datasets. Theysupportsyntactic parsing, information extraction, intent classification, and responsegeneration.Consequently, accurate POS tagging improves the overall reliability and naturalnessofacustomer support chatbot.

Conclusion
POS tagging is an important component of intelligent chatbot systems. Penn Treebanktagsprovide standardizedgrammaticalcategories,whileHMMtaggingusesprobabilitiestoresolvelexicalambigui ty. Contextandprobabilisticinformationcorrectlyidentify“book”asaverbinanimperativebookingreques t and as a noun when referring to an object.

CASESTUDY3:NEWSANALYTICSANDPOSTAGCORRECTIONSYSTEM

Introduction
Transformation-Based Tagging, commonly associated with the Brill Tagger, improvesaninitial POS-tagged sequence by applying learned transformation rules. Instead of acceptingtheinitial tagger's predictions directly, the system identifies systematic errors and correctsthemaccording to contextual patterns.
GivenSentence:
“Economicgrowthincreasesemployment.”
InitialPOSTags:
economic/JJ
growth/NN
increases/NNS
employment/NN
TransformationRul
e:

ChangeNNStoVBZiftheprecedingwordistaggedasNN.

1. ApplicationofTransformationRule
Initialsequence:
economic/JJgrowth/NNincreases/NNSemployment/
NN The word “increases” is initially tagged as:
increases/NNS
Theprecedingwordis:
growth/NN
Accordingtothelearnedrule:
NNS→VBZiftheprecedingwordisNN
Therefore:
increases/NNS→increases/VBZ
Corrected POS Sequence
economic/JJgrowth/NNincreases/VBZemployment/N N Linguistic Explanation
Inthesentence:
“Economicgrowthincreasesemployment.”
“growth”isthesubjectand“increases”expressestheactionperformedbythesubject. Therefore, “increases” should be a verb rather than a plural noun.

Since“growth”isathird-personsingularnoun,thepresent-tenseverb“increases”receivesthePenn Treebank tag:
VBZ– Verb,third-person singularpresent
Thus,thetransformation islinguisticallyappropriate.

2. AnalysisofInitialTaggingError
Theinitialtaggerproduced:
economic/JJgrowth/NNincreases/NNSemployment/
NN The main error is:
increases/NNS
ThetagNNSrepresents apluralnoun.
Thewordending“-s”mayhavecausedtheinitialsystemtointerpret“increases”asaplural noun.
However,itsgrammaticalroleinthissentenceisa verb.

Considerthesentencestructure:
Economicgrowth|increases|employment
Subject → Economic growth
Verb → increases
Object→employment
Therefore,thecorrectedsequenceis:
economic/JJgrowth/NNincreases/VBZemployment/N N The assignments are justified as follows:
economic/JJ:“economic”describesthenoun“growth”,soitfunctionsasanadjective. growth/NN: “growth” is a singular noun and functions as the subject. increases/VBZ: “increases” is a third-person singular present-tense verb. employment/NN:“employment”isanounandfunctionsastheobjectoftheverb.
ThisdemonstrateswhygrammaticalcontextisimportantwhenresolvingPOS-taggingambiguity.

3.WordFrequencyDistribution
Givencorpusfrequencies:
economic=120
growth=450
increases=210
employment=380
Step1:CalculateTotalFrequency
Total = 120 + 450 + 210 + 380
Total= 1160
Step2:CalculateRelativeFrequency
The formula is:
P(word)=Frequencyofword/Totalfrequency
Economic
P(economic)=120/1160
=0.1034 =10.34%
Growth
P(growth)=450/ 1160
=0.3879 =38.79%
Increases
P(increases)=210/1160
=0.1810 =18.10%
EmploymentP(employme
nt)=380/1160
=0.3276 =32.76%
FrequencyDistributionTable

Word	Frequ	RelativeFrequency	Per
economic		0.1034	10.34%
growth		0.3879	38.79%
increases		0.1810	18.10%
employm ent	380	0.3276	32.76%
Total	1160	1.0000	100%

RoleofFrequencyin POSTagging

Frequency information helps a probabilistic POS tagger estimate howcommonlywordsandword-tag combinations occur in a corpus.
A frequently observed word-tag combination can receive a higher probability thanararecombination.
However, word frequency alone cannot always determine the correct POStagbecause many words are ambiguous.

Therefore, effectiveprobabilisticPOStaggingcombines:
Wordfrequency+Tagfrequency+Context+Transitionprobabilities For example, even if “increases” appears frequently as a noun in some contexts, its positionafterthe singularnoun“growth”andbeforethenoun“employment”providesstronggrammaticalevidencethatitfunctions as a verb in the given sentence.

4.Transformation-BasedTaggingandEntropy

Transformation-Based Tagging
Transformation-BasedTaggingbeginswithaninitialPOS-taggedsequenceandthenapplieslearned correction rules.
Inthisexample:
Initial:
economic/JJgrowth/NNincreases/NNSemployment/
NN Rule:
ChangeNNStoVBZiftheprecedingwordisNN. Corrected:
economic/JJgrowth/NNincreases/VBZemployment/NN
Thetransformationcorrectsthegrammaticalroleof“increases”usingcontextualinformation.

Advantages
Transformation-BasedTagging:
•Correctssystematicmistakesproducedbytheinitialtagger.
•Usescontextualinformationfromneighboring wordsandtags.
•Producesinterpretablecorrectionrules.

•Combinesaninitialstatisticalorlexicaltaggingstagewithrule-based refinement.
•CanimproveoverallPOS taggingaccuracywhenusefultransformationruleshavebeenlearnedfromtraining data.

EntropyandTaggingUncertainty

Entropymeasuresuncertaintyinaprobabilitydistribution.
The general formula is:

$$H(X) = -\Sigma P(x) \log_2 P(x)$$

Supposeaninitialtaggerconsiderstwopossibletagsfor“increases”:
NNS and VBZ.
Iftheirprobabilitiesaresimilar,theentropywillberelativelyhighbecausethemodelisuncertain.
Forexample:
P(NNS)=0.50
P(VBZ)=0.50
Then:
$H=-[0.5 \log_2(0.5) +0.5\log_2(0.5)]$
H=1 bit

Thisrepresentsmaximumuncertaintybetweentwoequallylikelytags.
Afterapplyingastrongcontextualtransformationrule,thesystemmaybecomemoreconfidentinVBZ
. For illustration:
P(VBZ)=0.95
P(NNS)=0.05
Then:
$H=-[0.95 \log_2(0.95) +0.05\log_2(0.05)]$
H≈ 0.286 bits

Thus:
Beforecorrection→Higherentropy→Higheruncertainty After correction → Lower entropy → Higher confidence

Theexactbefore-and-afterentropycannotbecalculatedfromthecase-studydatabecausetagprobabilities are not supplied. The values above are illustrativeexamplesshowing how entropy can be used.

Evaluation

Transformation-Based Tagging improves the initial tagging stage by identifyingsystematicerrors and applying context-sensitive corrections. In the given sentence, theinitial NNSassignment is corrected to VBZ based on the preceding NN tag.
Entropycancomplementthisprocessbymeasuringtheuncertaintyassociatedwithcompeting POStags.
Areduction in entropy aftera reliable correction indicatesthat thesystemhas becomemoreconfident in its final tag assignment.

Conclusion

Transformation-Based Tagging provides an effective mechanismfor correctingerrorsgenerated during initial POS tagging. In the given example, the Brill-style transformationcorrectly changes “increases/NNS” to “increases/VBZ”. Corpus frequency informationsupports probabilistic decisions, while entropy provides a quantitative measure of uncertainty.Together, contextual rules, frequency analysis, and uncertainty measurement canimprovetheaccuracy of large-scale news analytics systems.

OVERALLCONCLUSION

Thethreecasestudiesdemonstrateimportant NaturalLanguageProcessingtechniquesusedinreal-world AI applications. N-gram language models, backoff, interpolation, and entropysupportintelligent word prediction in mobile keyboards. Penn Treebank POS tags and HiddenMarkovModels help chatbots identifygrammaticalrolesandresolvelexicalambiguity.Transformation-BasedTaggingimprovesinitial POS taggingresultsbyapplyinglearnedcontextualcorrections.TheseTechniquesenableNLPsystemstohandleuncertainty,sparsedata,ambiguouswords, andcomplexgrammaticalstructuresmoreeffectively.

RubricsforCalculation

Criteria	Weight	Level 4 Exemplary	Leve Proficient	
Under standi ng Case	25	Thorough understanding; clearlyidentif ies all key	Goodunderst ding withmostissu identified issues	
Applicati onof Concept s	25	Effectively appliesrelevant concepts analyzethec asedeeply	Applies appropriate conceptswit minorgaps	– apply
Analysis		In-depthanaly siswith reasoning insights	Goodanalys str with and reasoning	or no no

Solution Design	20	Innovative, feasiblesoluti onswith justification	Logicalsc adequa justification
Clarity		Clear, well-structured, and easy understand	Organized understandab t presentation

Q.No./ Criteria Level	Understandin gof Case	Application ofConcepts	Analysis	Soluti Design	Clarity	Total	Total Sub. %
1							
2							
3							

FacultyIn-charge	CourseCoordinator	ProgramDirector
Signature:	Signature:	Signature:
Date:	Date:	Date: